

Modern Generative AI (GenAI) Syllabus

1. Foundations of AI & Generative AI

- What is Artificial Intelligence
- Machine Learning vs Deep Learning vs Generative AI
- Types of generative models (Autoregressive, Diffusion, GAN, VAE)
- History of Generative AI
- Applications of AI across industries

2. Mathematics for AI

- Linear algebra
- Vectors and matrices
- Probability distributions
- Gradient descent
- Backpropagation
- Optimization algorithms

3. Programming for AI

- Python fundamentals
- NumPy and Pandas
- Data visualization
- Jupyter notebooks
- API usage

4. Machine Learning Fundamentals

- Supervised learning
- Unsupervised learning
- Regression and classification
- Clustering techniques
- Model evaluation metrics

5. Deep Learning Foundations

- Artificial Neural Networks
- CNN architectures
- RNN and LSTM

- Attention mechanisms
- Transformer architecture

6. Generative Models

- Generative modeling concepts
- GANs
- Variational Autoencoders (VAE)
- Diffusion models
- Autoregressive models

7. Large Language Models

- Tokenization
- Embeddings
- Transformer-based LLM architecture
- Pretraining and fine-tuning
- RLHF (Reinforcement Learning with Human Feedback)

8. Prompt Engineering

- Prompt design
- Zero-shot and few-shot prompting
- Chain-of-thought prompting
- Role prompting
- Prompt templates and guardrails

9. Building GenAI Applications

- Chatbots
- Document Q&A; systems
- AI copilots
- Content generation systems

10. Retrieval Augmented Generation (RAG)

- Embeddings and vector search
- Chunking and indexing
- Vector databases
- Hybrid retrieval systems

11. Agentic AI Systems

- AI agents
- Planning and reasoning
- Tool usage
- Multi-agent collaboration
- Memory systems

12. Multimodal AI

- Text-to-image models
- Image-to-text models
- Speech models
- Video AI
- Vision-language models

13. Fine-Tuning and Model Customization

- LoRA and QLoRA
- Instruction tuning
- Domain adaptation

14. Evaluation of GenAI Systems

- Hallucination detection
- Evaluation metrics
- Human evaluation
- Safety and guardrails

15. MLOps & LLMOps

- Model deployment
- Monitoring and observability
- Model versioning
- Continuous training

16. Responsible AI

- Bias and fairness
- AI governance
- Privacy
- Responsible AI frameworks

17. Production-Grade GenAI Systems

- Scalable GenAI architecture
- Cost optimization
- Security considerations

18. Real-World Projects

- AI document assistant
- Customer support chatbot
- AI DevOps assistant
- Multimodal chatbot
- Autonomous AI agent